

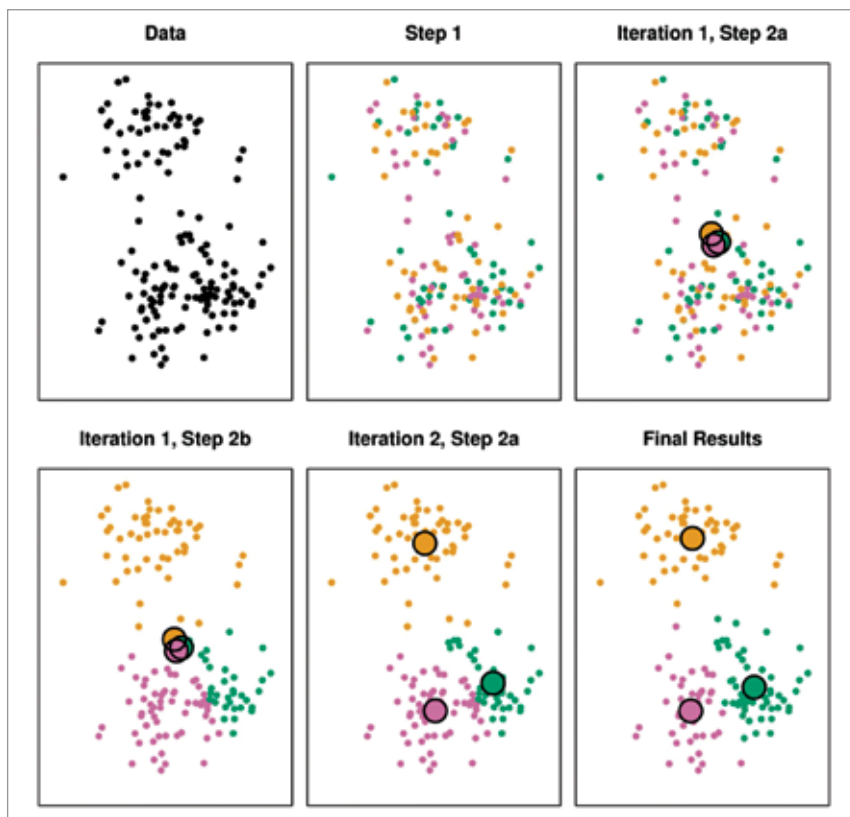


Fig. 3: K-means clustering algorithm (Courtesy: Google Images)

Because a pedestrian does not move on the road the same way as a car does. Nor does a deer move the same way a cow does. This is important information because only by knowing what kind of an obstacle you are dealing with can you deal with it properly. A deer can be fidgety and make sudden moves. A cow may not care as much about being run over. A person could be jay-walking and a car might suddenly shift lanes without any signal or warning sign.

This is where lidar technology comes in. Unlike its predecessor, lidar fires a train of super-short laser pulses to get the depth resolution of the object. In order for these laser pulses to be shot accurately at regular intervals, we use what is called an integrated photonics circuit to periodically block the light.

The data of the Internet is carried by precision timed pulses of light shot every 100 picoseconds (0.0000000001 second). This is done with the help of what is called a 'Mach-Zehnder modulator' that uses wave interference to achieve this goal. The light pulse lasting 100 picoseconds leads to a depth

resolution of a few centimeters.

Apart from radars and lidars, high-tech cameras are also used to identify and understand traffic lights and measure distance between different objects on the road. They are also used to cover potential blind spots.

Algorithms used in the vehicles

The algorithms used in self-driving vehicles belong to the family of what we refer to as 'machine learning' algorithms. But that is a broad term. Machine learning can mean anything from forecasting the weather based on historical weather data to recognising patterns in transactions made by customers at a bank. In order to bring more specificity, we need to briefly discuss what machine learning is.

Machine learning can be loosely defined as the ability of a machine (primarily a computer) to accomplish a task effectively by learning through repetitions. This is similar to how humans learn. We act based on some preliminary knowledge, see the results, contemplate on whether those results were what we were looking for, and then reassess our actions accord-

ingly. Over time, we keep getting better at performing our tasks because we are learning from our mistakes.

Similarly, machines can also be trained to perform certain tasks more efficiently by implementing what we call 'machine learning' algorithms. Machine learning algorithms can be broadly divided into three types, namely, supervised learning, unsupervised learning, and reinforcement learning.

Supervised learning type of algorithms work in conjunction with datasets. Basically, a computer is provided with (ideally) large sets of data and supervised learning algorithms are designed to train the computer based on that data. The better the training algorithm, the lesser the probability of error by the machine. Linear regression is a good example of supervised learning, which is often used for tasks such as predicting the probability of changes in the weather given the historical data.

Unsupervised learning type of algorithms work in conjunction with 'untagged' datasets. Not all datasets provide you with a clear picture and you may not always know what you are looking for. In such scenarios, unsupervised learning algorithms are used on these datasets to help find certain patterns. These algorithms also learn and get better at reducing the probability of errors with each iteration. Clustering algorithms such as K-means are a good example of unsupervised learning algorithms that help find patterns in large datasets.

Reinforcement learning type of algorithms are the most similar in terms of how humans learn. They consist of three major elements, namely, a goal, a utility function, and a reward system. A machine is placed in an environment with a predefined goal or set of goals. Rewards are set for each action based on the goal and the utility function is given to the machine. The utility function helps the machine take the actions that have the highest probability of reward and the lowest probability of risk. The focus is to reach the goal by gaining the highest rewards possible.

So, what type of machine learning algorithms do autonomous vehicles utilise? The answer, all of the above.